ScholarPlan: An LLM-Powered Planning Agent for Academic Research and Information Gathering

Design Proposal Report

Module: Intelligent Agents

Group A

September 2026

1. Introduction

Academic literature searching is repetitive yet error-prone when delegated to a single large language model (LLM), because unaided LLMs plan unreliably and fabricate citations (Ji *et al.*, 2023; Kambhampati *et al.*, 2024). This proposal presents ScholarPlan, a multi-agent system in which an LLM Planner decomposes a research goal, cooperating agents retrieve, summarise, and verify evidence from open scholarly sources, and an Archivist stores the results offline.

2. System Requirements and Assumptions

2.1 Requirements

- **Instruction-tuned LLMs for three roles.** The Planner and Processor must return schema-valid JSON (a subtask graph and a claim list) and are ranked on the Berkeley Function-Calling Leaderboard (BFCL, 2026); the Critic performs entailment classification and is ranked on LLM-AggreFact (Tang, Laban and Durrett, 2024).
- **Keyless scholarly APIs** (arXiv, OpenAlex, Crossref) that return stable identifiers (DOI, arXiv ID), so every claim is traceable to a source (Priem, Piwowar and Orr, 2022).
- **Local sentence embeddings** (Reimers and Gurevych, 2019) for deduplication, relevance ranking, and vector indexing at zero cost.
- **A bounded ReAct reasoning loop** (Yao *et al.*, 2023) capped at 20 steps per run, controlling both Azure credit and API rate limits.
- **Offline outputs:** a report, BibTeX bibliography, vector index, and execution trace, satisfying the retrieval, processing, and storage success criteria.

2.2 Assumptions

- Sources are mainly English; full text exists only for open-access works, with abstracts as fallback.
- Single user, single run, single machine; no authentication, multi-tenancy, or distributed deployment.
- Free-tier quotas and the one-off Azure student credit suffice for demonstration scale.

3. Design Decisions

- **Hybrid agent architecture.** A single deliberative LLM Planner reasons over the goal while role-specialised reactive agents (Retriever, Processor, Critic, Archivist) act only when their input appears. Hybrid layering combines deliberative goal-direction with reactive robustness (Wooldridge, 2009) and keeps retrieval, processing, and critique as separate failure surfaces, so an irrelevant source is never confused with an unsupported claim (Guo *et al.*, 2024).
- **Blackboard coordination.** Agents never invoke one another; they read and write shared state (Figures 2 and 3). The blackboard model offers decoupled, opportunistic control among heterogeneous knowledge sources (Hayes-Roth, 1985) and has recently outperformed centralised orchestration for LLM agents in information discovery (Salemi *et al.*, 2025). Implementing it in SQLite3 makes every intermediate artefact persistent and inspectable.
- **ReAct planning with Reflexion-style re-planning.** The Planner interleaves Thought, Action, and Observation (Yao *et al.*, 2023). A rejected summary produces a written Feedback record that the Planner reads before its next thought, then narrows the query, seeks another source, or abandons the claim, mirroring verbal self-reflection (Shinn *et al.*, 2023).
- **Independent verification.** The Critic, a separate model, tests whether each claim is entailed by its cited source, following the LLM-Modulo argument that LLM plans need an external verifier (Kambhampati *et al.*, 2024) and grounding outputs in retrieved text (Lewis *et al.*, 2020).
- **Benchmark-driven model selection.** GPT-5-mini serves the Planner and Processor: it scores 55.46% on BFCL V4, and Azure's $100 student credit funds well over a hundred runs at Global Standard prices (BFCL, 2026; Microsoft, 2026), with GPT-5-nano as fallback. Llama-3.3-70B-Instruct (Dubey et al., 2024) serves as the Critic: it outscores gpt-4o-mini on LLM-AggreFact (Tang, Laban and Durrett, 2024), is not deprecated in Azure, and, being open-weight, can run locally via Ollama for reproducibility.
- **Plain-Python orchestration.** LangChain and LangGraph were rejected because their abstractions obscure the ReAct loop under assessment and add dependency churn.

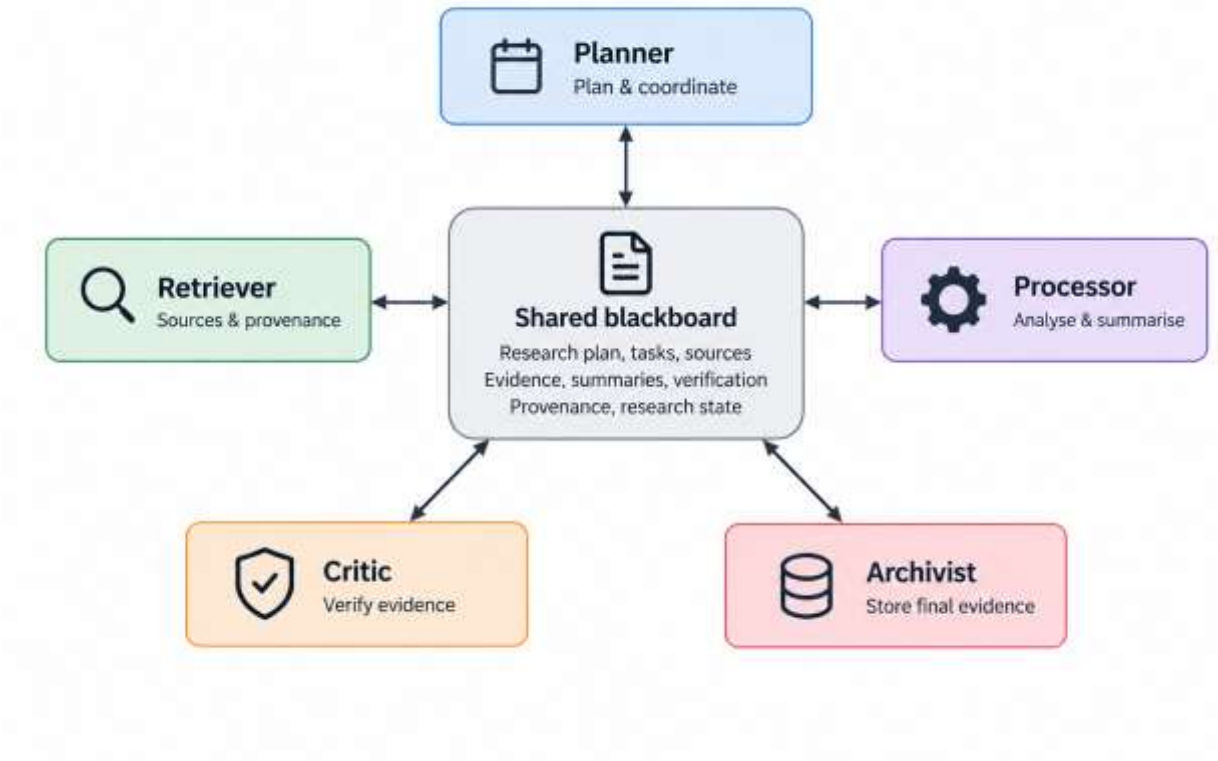


Figure 1. Architectural view: five role-specialised agents coordinating through the shared blackboard.

4. Development Approach

- **Scrum with weekly sprints** (Schwaber and Sutherland, 2020). LLM behaviour is empirically uncertain, so requirements emerge through experimentation; plan-driven methods presuppose stable, specifiable requirements (Sommerville, 2016). Sprint reviews double as prompt-evaluation checkpoints.
- **Role-based agent-oriented analysis.** Following the Gaia methodology, each agent is specified as a role with responsibilities, permissions (blackboard read and write rights), and protocols (Wooldridge, Jennings and Kinny, 2000); the five roles map onto per-member work packages, supporting virtual teamwork.
- **Testing by agent type.** Deterministic agents (Retriever, Archivist) receive pytest unit tests; the Critic is unit-tested against bespoke-minicheck:7b; the Planner and Processor use an evaluation harness of fixed research questions with known relevant sources, measuring source recall and Critic pass rate.

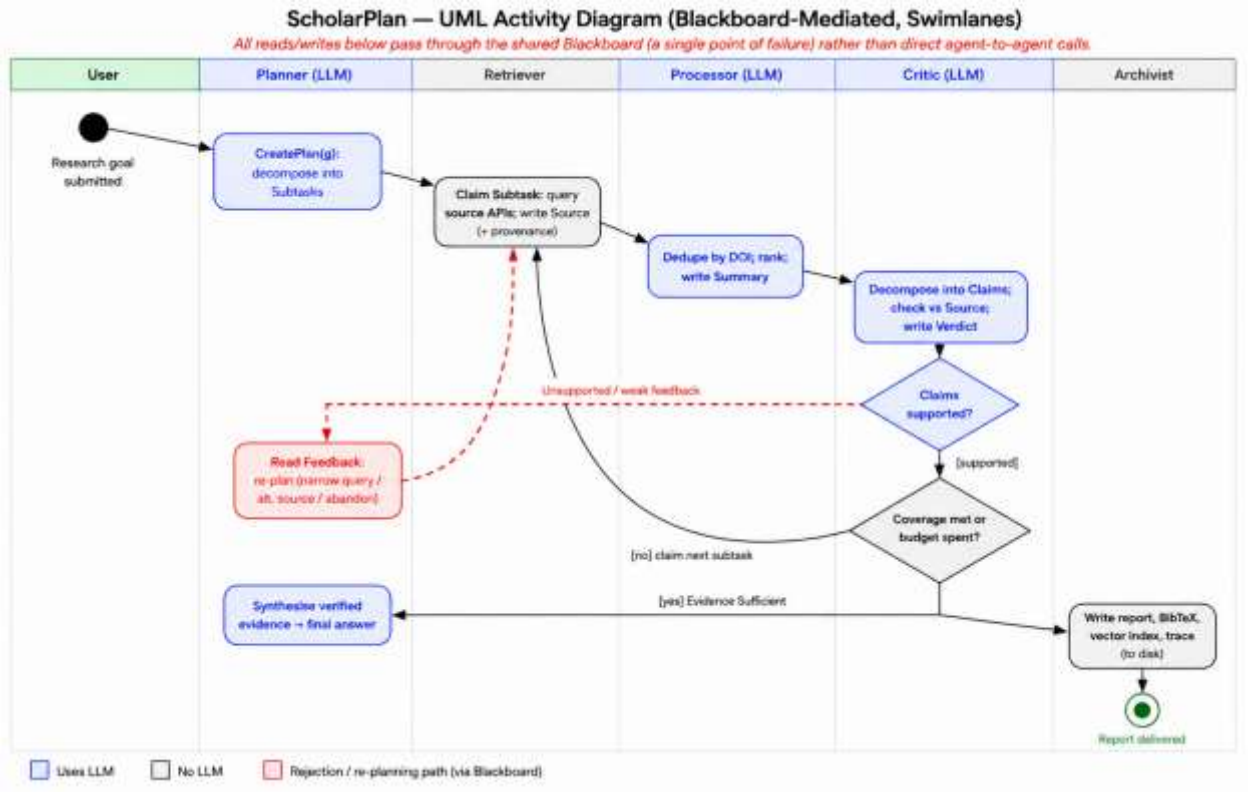


Figure 2. Activity view: end-to-end flow by agent, with the Critic-rejection re-planning loop dashed.

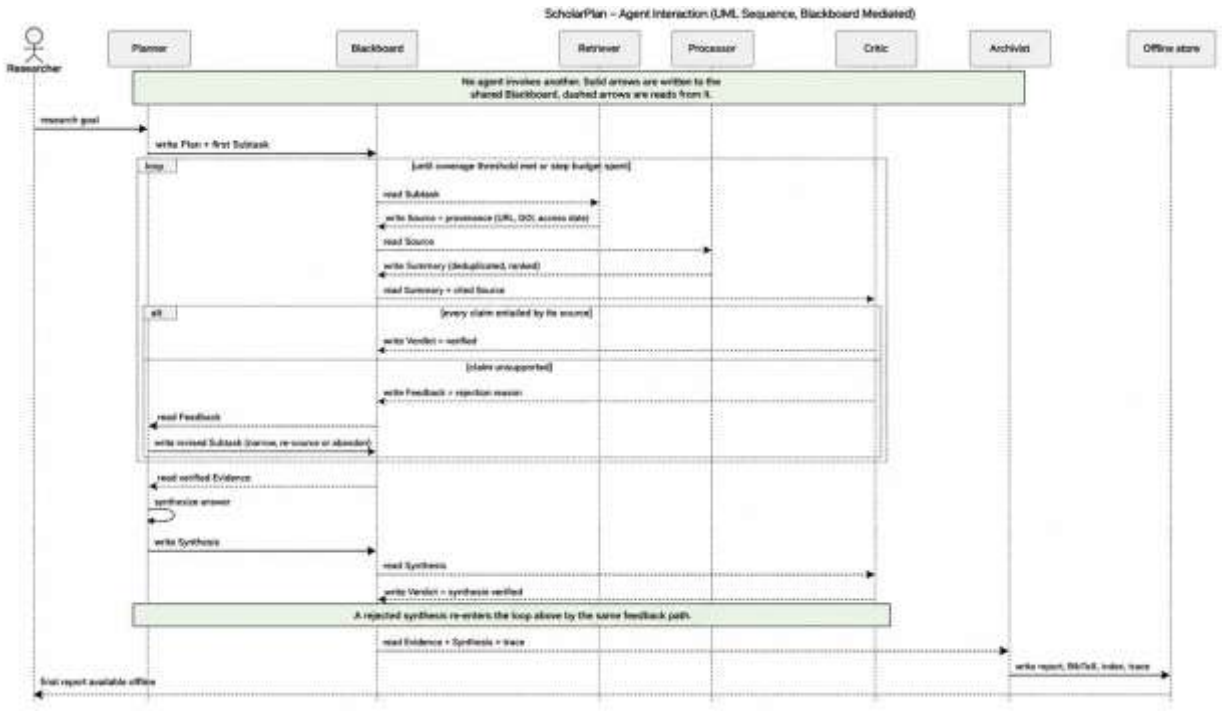


Figure 3. Sequence view: agents interacting through the blackboard; only the Archivist writes to disk.

5. Anticipated Challenges

- **Task decomposition.** LLM planners over- or under-decompose goals (Huang *et al.*, 2024). Mitigation: JSON-schema-constrained subtasks, one subtask issued per ReAct step, and a coverage threshold that terminates the loop, keeping plan validity under external control (Kambhampati *et al.*, 2024).
- **Unreliable or noisy online data.** Irrelevant context degrades LLM reasoning (Shi *et al.*, 2023). Mitigation: DOI deduplication, embedding-based relevance ranking (Reimers and Gurevych, 2019), and provenance captured at fetch time (Priem, Piwowar and Orr, 2022).
- **Hallucinated claims.** Summaries may assert content absent from the source (Ji *et al.*, 2023). Mitigation: claim-level entailment checking by the Critic (Tang, Laban and Durrett, 2024); unsupported summaries never reach the report.
- **Explainability of LLM decisions.** Mitigation: every Thought, Verdict, and Feedback is persisted to the blackboard and exported as a trace, making the reasoning chain auditable in the manner of chain-of-thought (Wei *et al.*, 2022).
- **Agent cooperation and single point of failure.** The blackboard decouples agents but concentrates risk (Salemi *et al.*, 2025). Mitigation: SQLite transactions for atomic claims and the step budget to prevent runaway loops.
- **Cost and rate limits.** Azure credit is non-renewable, and arXiv permits one request every three seconds. Mitigation: the 20-step budget, source caching, and an early sprint measuring real per-run cost, since benchmark costs cannot simply be summed.
- **Non-deterministic behaviour.** Mitigation: the evaluation harness in Section 4 and low-temperature sampling.

6. Tools, Libraries, Models and Environments

- **Language and environment:** Python 3.11; uv for virtual environments and dependencies; VS Code; Git and GitHub for source control and code review.
- **Models:** GPT-5-mini (Planner, Processor; fallback GPT-5-nano) and Llama-3.3-70B-Instruct (Critic) on Azure for Students; bespoke-minicheck:7b via Ollama for testing; sentence-transformers for embeddings.
- **Data sources:** arXiv API, OpenAlex API, and Crossref REST API.
- **Storage:** sqlite3 (blackboard, metadata) and sqlite-vec (vector index).
- **Testing:** pytest and a bespoke evaluation harness.

- **Hosting:** Azure for Students ($100 credit) was selected over Hugging Face inference credits ($0.10 per month, too small for repeated evaluation) and Ollama (hardware the team cannot guarantee).

7. Conclusion

ScholarPlan combines a deliberative ReAct planner, reactive specialists, and blackboard coordination to meet the three success criteria within free-tier constraints: retrieval by the Retriever, processing by the Processor and Critic, storage by the Archivist. Its principal risks (decomposition quality, noisy data and cost) are addressed through schema-constrained planning, independent verification and a bounded step budget.

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